Topic 7 Review [98 marks]

1. [Maximum mark: 15]

Each set of traffic lights in a city is controlled by a separate embedded system.

(a.i) Describe what is meant by an embedded system. [2]

(a.ii) State **two** other examples of embedded systems in daily life. [2]

It has been proposed that all the traffic lights in the city should be controlled from a central computer.

(b) Discuss the advantages **and** disadvantages of using one central computer to manage the city's traffic light system. [4]

Vehicle counts provide the city's traffic control centre with information regarding the number of vehicles on the city's roads.

(c) List **two** types of sensor that could be used to collect data on the number of vehicles. [2]

(d) Describe **one** difference between interrupts and polling. [2]

The data about traffic and road conditions will be shared with application (app) developers.
Their apps can provide real-time information about traffic conditions.

(e) Explain the benefit for drivers of having access to real-time information about traffic and road conditions. [3]

2. [Maximum mark: 2]

Outline **one** advantage of using robots in the manufacturing of cars. [2]

3. [Maximum mark: 2]
A temperature sensor is used in an automatic washing machine to help maintain the temperature of the water.

Outline the use of **one other** type of sensor used in an automatic washing machine. [2]

4. [Maximum mark: 4]
Describe the role of feedback in a system that uses sensors and a microprocessor to control the temperature of a room. [4]

5. [Maximum mark: 15]
Input devices that detect cars approaching a crossroads are connected to a microprocessor.

(a.i) Identify **two** types of sensor that can be used to detect approaching cars. [2]

(a.ii) Outline why sensors are appropriate input devices in this situation. [2]

(b) Suggest the type of memory that could be used to store the control program in the microprocessor. [2]

The traffic lights at the crossroads are also connected to a microprocessor. A person who wishes to cross the road presses a button at a traffic light. This causes an interrupt.

(c.i) Outline what is meant by an interrupt. [2]

(c.ii) Explain how the microprocessor can deal with this interrupt. [3]

Cameras are installed on the top of the traffic lights at the crossroads.

(d.i) Outline **one** benefit of monitoring the traffic with cameras. [2]

(d.ii) Outline **one** concern about monitoring the traffic with cameras. [2]

6. [Maximum mark: 15]

Smart control systems can manage the temperature within a house.

(a) Outline the steps involving the sensor, processor and output transducer to manage the temperature in the house. [5]

(b) Describe the role of feedback in this control system. [2]

(c) The smart control system is managed by an operating system.

(c.i) Describe **one** function of an operating system. [2]

(c.ii) Outline **one** reason why a dedicated operating system would be used. [2]

(d) Compare and contrast a centralized control system with a distributed control system for managing the temperature of a house. [4]

7. [Maximum mark: 15]

A company is planning to transform its office building into a smart building. Among other things, a smart building can control the opening and closing of its doors.

(a) Outline **two** other operations that can be controlled by a smart building. [4]

Sensors, processors and output transducers are vital components of a smart building. They play an important role in the collection and management of data.

(b) Explain how a smart building can control the opening and closing of all its doors. You should refer to sensors, processors and output transducers. [4]

An operating system has a significant role in a smart building system.

(c.i) Identify **two** functions of this operating system. [2]

(c.ii) Suggest **one** technique this operating system might use to determine when a hardware device needs attention. [2]

(d) Explain why transforming the building into a smart building will be beneficial for the company. [3]

8. [Maximum mark: 15]

Cars have many automated features to improve the driving experience. One example is the use of headlights that automatically switch on and off.

(a) State **one** input device used by the automated headlights. [1]

Some cars have adaptive cruise control. This uses RADAR technology and a processor to ensure the car stays the same distance away from the car in front.

(b) Explain how the interaction between the inputs, processing and the outputs of the feedback loop ensure the car stays the same distance away from the car in front. [6]

A system with many functions operating independently could be managed using either centralized or decentralized processing.

(c) Discuss the use of centralized and decentralized processing in the context of control systems such as those involved in the operation of motor vehicles. [5]

As vehicles become fully autonomous (self-driving), ethical considerations must be taken into account when designing software.

(d) Explain **one** ethical concern that must be considered in the development of autonomous vehicles. [3]

9. [Maximum mark: 15]

A rail transport company uses a global positioning system (GPS) to determine a train's position.

(a) Explain how GPS works. [4]

The GPS data is used to provide real-time arrival data on video displays in train stations.

(b.i) State **one** data item concerning the arrival of a train that may be provided on the video display. [1]

(b.ii) Outline how sensors can be used in combination with GPS to provide more accurate arrival data. [2]

The GPS data is made publicly available so that software developers can use it to build apps.

The mobile application *ATrainAway* uses the real-time train GPS data as well as the GPS data from the user's smartphone.

(c) Outline **two** pieces of information that the *ATrainAway* application could provide to the user. [4]

(d) Discuss the advantages **and** disadvantages of using GPS in transportation systems. [4]